# Geodesic Pathwise Polynomial Annihilators on Graphs

`gflow` graph trend filtering project notes

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**Abstract**

This note develops a graph-native version of polynomial trend filtering based on geodesic paths. The motivating question is simple: in one dimension, degree-$r$ polynomials are exactly the functions annihilated by an $(r+1)$-st derivative or divided-difference operator. Can a graph operator have an analogous null space, without first embedding the graph in Euclidean coordinates? The proposed answer is to sample geodesic paths through graph disks, fit or construct one-dimensional polynomial annihilators along those paths, and assemble the resulting rows into a global penalty matrix. The core object is an $(r+1)$-st pathwise polynomial annihilator over the composite geodesic path family $\Gamma(v, \varepsilon)$.

The note also discusses a practical complication that appears on general graphs: the local polynomial dimension need not be known in advance. We therefore describe local intrinsic-dimension estimation, smoothing of those estimates over the graph, and their use in choosing how many local annihilator rows should be retained.

## 1 The Question

In ordinary one-dimensional trend filtering, the degree of the unpenalized polynomial space is controlled by the order of the difference operator. For example,

$$D^{(3)} f = 0 \quad \Longleftrightarrow \quad f \text{ is quadratic}$$

on a connected interval, up to the usual discrete boundary conventions.

For a smooth function $f : \mathbb{R}^n \to \mathbb{R}$, the direct continuum analogue is also clear:

$$\nabla^{r+1} f = 0 \quad \Longleftrightarrow \quad f \text{ is a polynomial of total degree at most } r$$

on a connected open set. Here $\nabla^{r+1} f$ denotes the full tensor of all $(r+1)$-st partial derivatives.

Graphs do not have coordinate axes or partial derivatives by default. They do, however, have paths and geodesic distance. The purpose of this document is to turn that observation into a concrete operator.

## 2 A Guiding Smooth Fact

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is smooth near the origin. If for every line through the origin the restriction

$$t \mapsto f(tv)$$

is a polynomial of degree at most $r$, with a uniform degree bound over all directions $v$, then $f$ is a multivariate polynomial of total degree at most $r$.

This fact is the key bridge. It says that testing polynomial behavior along all lines through a point is not merely a one-dimensional trick. Smoothness forces the directional polynomial data to assemble into a true multivariate polynomial.

For graphs, geodesics through a vertex are the intrinsic analogue of lines through a point. This suggests the following principle:

> *A graph function is locally degree-$r$ polynomial near $v$ if it looks degree-$r$ polynomial along geodesics through $v$.*

# 3 Composite Geodesic Paths

Let

$$G = (V, E, \ell)$$

be an undirected graph with positive edge lengths $\ell_e$. Let

$$d_G(u, w)$$

denote shortest-path distance. For a vertex $v$ and radius $\varepsilon > 0$, define the graph disk

$$D(v, \varepsilon) = \{u \in V : d_G(u, v) \le \varepsilon\}.$$

The boundary of the disk is

$$\partial D(v, \varepsilon) = \{u \in D(v, \varepsilon) : \text{some neighbor of } u \text{ lies outside } D(v, \varepsilon)\}.$$

For two boundary vertices $a, b \in \partial D(v, \varepsilon)$, choose shortest paths

$$\gamma_{a,v}, \qquad \gamma_{v,b}.$$

Their concatenation

$$\gamma_{a,b}^{(v)} = \gamma_{v,b} \circ \gamma_{a,v}$$

is a composite path through $v$. We retain it as a composite geodesic path when

$$d_G(a, b) = d_G(a, v) + d_G(v, b).$$

In words: the path from $a$ to $b$ through $v$ must itself be geodesic.

The set of such paths is denoted

$$\Gamma(v, \varepsilon).$$

# 4 Pathwise Polynomial Annihilators

Let

$$\gamma = (v_0, \dots, v_m) \in \Gamma(v, \varepsilon)$$

be a composite geodesic path containing $v = v_j$. Give this path a one-dimensional coordinate system centered at $v$:

$$t_j = 0,$$

$$t_i = -d_G(v_i, v) \quad \text{for } i < j, \qquad t_i = d_G(v_i, v) \quad \text{for } i > j.$$

Now a function $f : V \to \mathbb{R}$ restricted to $\gamma$ becomes one-dimensional data:

$$(t_i, f(v_i)), \qquad i = 0, \dots, m.$$

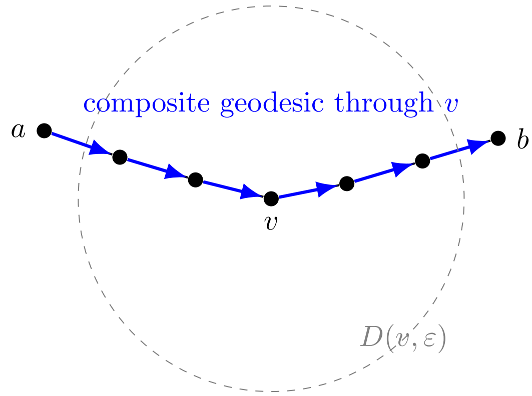


Figure 1: A composite geodesic path through $v$ is built by joining two shortest paths from $v$ to disk-boundary vertices. It is retained only when the joined path is itself shortest between its endpoints.

### 4.1 Exact Divided-Difference Rows

If we use exactly $r + 2$ points along a path, then the $(r + 1)$-st divided difference

$$[t_0, \dots, t_{r+1}]f$$

annihilates every degree-$r$ polynomial in $t$. This gives a row vector

$$a_\gamma^{(r+1)}$$

such that

$$a_\gamma^{(r+1)} f_\gamma = 0$$

whenever $f_\gamma$ is sampled from a polynomial of degree at most $r$ along the path.

### 4.2 Regression-Based Rows

For longer paths, it is often better to use all points on the path. Let

$$X_\gamma = \begin{bmatrix} 1 & t_0 & t_0^2 & \cdots & t_0^{r+1} \\ 1 & t_1 & t_1^2 & \cdots & t_1^{r+1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & t_m & t_m^2 & \cdots & t_m^{r+1} \end{bmatrix},$$

and let $W_\gamma$ be a diagonal kernel weight matrix centered at $t = 0$. The weighted least-squares coefficient vector is

$$\widehat{\alpha} = (X_\gamma^T W_\gamma X_\gamma)^{-1} X_\gamma^T W_\gamma f_\gamma.$$

The coefficient $\widehat{\alpha}_{r+1}$ is a linear functional of the path values:

$$\widehat{\alpha}_{r+1} = a_\gamma^{(r+1)} f_\gamma,$$

where

$$a_\gamma^{(r+1)} = e_{r+2}^T (X_\gamma^T W_\gamma X_\gamma)^{-1} X_\gamma^T W_\gamma.$$

This row also annihilates degree-$r$ polynomials, because the fitted $(r + 1)$-st coefficient is zero whenever the data lie exactly in the column space of polynomials of degree at most $r$.

## 5 The Graph Operator

For each vertex $v$, each composite geodesic $\gamma \in \Gamma(v, \varepsilon)$, and degree $r$, build a row $a_\gamma^{(r+1)}$. Assemble all rows into a global sparse matrix:

$$A_{r+1,\mathrm{geo}}.$$

The corresponding graph trend-filtering estimator is

$$\widehat{f}_\lambda = \arg\min_{f \in \mathbb{R}^{|V|}} \left\{ \frac{1}{2} \|y - f\|_2^2 + \lambda \|A_{r+1,\mathrm{geo}} f\|_1 \right\}.$$

The intended null-space property is

$$A_{r+1,\mathrm{geo}} f = 0 \quad \Longleftrightarrow \quad f \text{ is locally degree-}r\text{ polynomial along sampled geodesics.}$$

This is not automatically the same as a Euclidean multivariate polynomial space. It is a graph-intrinsic analogue. On sufficiently rich samples of a smooth manifold, however, this condition should approximate the smooth fact that degree-$r$ behavior along all lines through a point characterizes degree-$r$ polynomial behavior.

## 6 Estimating Local Intrinsic Dimension

The discussion above assumes that the local polynomial dimension is known. On a sampled square this is easy: the intrinsic dimension is two, so the dimension of degree-$r$ polynomials is

$$\dim \mathcal{P}_r(\mathbb{R}^2) = \binom{r+2}{2}.$$

On a general graph, however, there is no given coordinate dimension. A graph disk may look one-dimensional near a boundary, two-dimensional in a sampled surface region, or higher-dimensional near a highly branched or noisy part of the graph. Therefore a local row-selection rule should not require a fixed dimension as a hidden assumption.

Let $m_v$ denote the effective intrinsic dimension near vertex $v$. If $m_v$ were known, the local degree-$r$ polynomial dimension would be

$$p_v(r) = \binom{m_v + r}{r}.$$

For example,

| $m_v$ | $r=1$ | $r=2$ | $r=3$ |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 2 | 3 | 6 | 10 |
| 3 | 4 | 10 | 20 |

Thus the local dimension estimate directly determines the null-space size that rank-based row thinning should try to preserve.

### 6.1 Local MDS/PCA Dimension

A simple geometric estimator starts from pairwise graph-geodesic distances inside the disk $D(v,\varepsilon)$. Apply classical multidimensional scaling (MDS), or an equivalent local principal-coordinate analysis, to obtain nonnegative local eigenvalues

$$\lambda_{v,1} \geq \lambda_{v,2} \geq \cdots \geq 0.$$

The dimension can be estimated by an energy threshold:

$$\widehat{m}_v^{\mathrm{MDS}} = \min \left\{ m : \frac{\sum_{j=1}^{m} \lambda_{v,j}}{\sum_{j\geq 1} \lambda_{v,j}} \geq \alpha \right\},$$

with a default such as $\alpha = 0.90$ or $0.95$. An alternative is an eigengap rule:

$$\widehat{m}_v^{\mathrm{gap}} = \arg\max_{j} \frac{\lambda_{v,j}}{\lambda_{v,j+1} + \delta},$$

where $\delta > 0$ prevents division by zero.

The advantage of the MDS/PCA view is that it gives both an estimated dimension and a diagnostic spectrum. If the local spectrum has a clear drop after two components, then a two-dimensional polynomial target is well supported. If the spectrum decays slowly, then the graph disk does not provide a clean local Euclidean model.

### 6.2 Volume-Growth Dimension

A more graph-native estimator uses the growth rate of balls. For radii

$$0 < \rho_1 < \rho_2 < \cdots < \rho_L \leq \varepsilon,$$

compute

$$N_v(\rho_\ell) = |B_G(v, \rho_\ell)|.$$

On an $m$-dimensional smooth space, one expects locally

$$N_v(\rho) \approx C_v \rho^m.$$

Taking logs gives

$$\log N_v(\rho) \approx \log C_v + m \log \rho.$$

Thus one can estimate

$$\widehat{m}_v^{\mathrm{growth}} = \text{slope of a local regression of } \log N_v(\rho) \text{ on } \log \rho.$$

This estimator is intrinsic and does not require an embedding, but it is often noisy on small graph disks because vertex counts are discrete and boundary effects are strong.

### 6.3 Doubling Dimension

A related estimate compares ball counts at two radii:

$$\widehat{m}_v^{\mathrm{dbl}}(\rho) = \log_2 \frac{|B_G(v, 2\rho)|}{|B_G(v, \rho)|}.$$

If the graph locally resembles an $m$-dimensional space, this ratio should be near $m$. Doubling estimates are easy to compute and easy to interpret, but they can be unstable when $B_G(v,\rho)$ is very small or when $2\rho$ is already close to the disk boundary.

### 6.4 Combining And Smoothing Dimension Estimates

No single estimator should be trusted blindly. A practical implementation can compute several raw estimates:

$$\widehat{m}_v^{\mathrm{MDS}}, \qquad \widehat{m}_v^{\mathrm{growth}}, \qquad \widehat{m}_v^{\mathrm{dbl}}.$$

These can be combined by a median, a trimmed mean, or a confidence-weighted average:

$$\widehat{m}_v^{\mathrm{raw}} = \mathrm{Combine}\left(\widehat{m}_v^{\mathrm{MDS}}, \widehat{m}_v^{\mathrm{growth}}, \widehat{m}_v^{\mathrm{dbl}}\right).$$

For example, the MDS estimate could receive high confidence when the eigengap is large, while the growth estimate could receive low confidence when the available radius range is narrow.

Because intrinsic dimension should usually vary slowly across a sampled manifold-like graph, the raw estimates should then be smoothed over the graph. One may use graph low-pass smoothing, local median smoothing, or kernel regression over graph distance:

$$\widetilde{m}_v = \frac{\sum_{u \in B_G(v,h)} K(d_G(u,v)/h)\, \widehat{m}_u^{\mathrm{raw}}}{\sum_{u \in B_G(v,h)} K(d_G(u,v)/h)}.$$

The final integer dimension can then be obtained by rounding or thresholding:

$$m_v^\star = \min\{m_{\max}, \max\{1, \mathrm{round}(\widetilde{m}_v)\}\}.$$

This smoothing step is important. Without it, neighboring vertices may alternate between dimensions 1, 2, and 3 because of small changes in local sampling density. Such flickering would make the row-selection rule unstable.

### 6.5 Use In Rank-Based Row Thinning

Let

$$n_v = |D(v, \varepsilon_v)|$$

be the number of vertices in the local disk. Once a smoothed dimension estimate $m_v^\star$ is available, the local polynomial dimension target is

$$p_v(r) = \binom{m_v^\star + r}{r}.$$

If $A_v^{\mathrm{cand}}$ is the matrix of all candidate annihilator rows for the disk, then a rank-thinning rule can aim for local rank

$$\mathrm{rank}(A_v) \approx n_v - p_v(r).$$

Equivalently, the local nullity target is

$$\dim \mathrm{null}(A_v) \approx p_v(r).$$

This suggests a constructive thinning algorithm:

1. Generate candidate pathwise annihilator rows in $D(v, \varepsilon_v)$.
2. Estimate $m_v^\star$ from local graph geometry.
3. Set the target local rank to $n_v - \binom{m_v^\star + r}{r}$.

4. Add candidate rows greedily only when they increase numerical rank or improve conditioning.

5. Stop when the target rank is reached, or when all useful candidate rows have been exhausted.

The dimension estimate should guide row thinning, not certify success. The operator still needs to be tested by numerical nullity and polynomial residuals:

$$\widehat{\dim \mathrm{null}}(A) \quad \text{and} \quad \frac{\|Af_{\mathrm{poly}}\|_2}{\|f_{\mathrm{poly}}\|_2}.$$

The purpose of estimating local dimension is to prevent row thinning from silently imposing a two-dimensional polynomial target on graph regions whose geometry is effectively one-dimensional, three-dimensional, branched, or too irregular for a clean polynomial model.

### 6.6 Using Local Nullity And Polynomial Residuals During Thinning

The local dimension estimate is not the only information available during row selection. Once candidate rows have been generated on a disk, we can also measure directly whether the selected rows preserve the intended polynomial space. The important distinction is local versus global use.

Let $P_v$ be a local polynomial design matrix on $D(v, \varepsilon_v)$. If local coordinates $z_u \in \mathbb{R}^{m_v^\star}$ have been estimated for vertices

$$u \in D(v, \varepsilon_v),$$

then $P_v$ contains all monomials up to degree $r$ in those local coordinates. For example, if $m_v^\star = 2$ and $r = 2$, then the columns are

$$1, \quad z_1, \quad z_2, \quad z_1^2, \quad z_1 z_2, \quad z_2^2.$$

For a candidate selected local row matrix $A_{v,S}$, define the local polynomial residual

$$\rho_v(S) = \frac{\|A_{v,S} P_v\|_F}{\|P_v\|_F}.$$

Also define the local nullity estimate

$$\nu_v(S) = \dim \mathrm{null}(A_{v,S}).$$

The desired target is

$$\nu_v(S) \approx p_v(r) = \binom{m_v^\star + r}{r}, \qquad \rho_v(S) \approx 0.$$

Thus local nullity and local polynomial residuals can be used constructively inside the thinning algorithm. A useful row-selection objective is:

$$\min_{S \subseteq \mathcal{R}_v} \left\{ \rho_v(S) + \mu \, |\nu_v(S) - p_v(r)| + \eta \, \kappa_v(S) + \zeta \, C_v(S) \right\},$$

where $\mathcal{R}_v$ is the candidate row set, $\kappa_v(S)$ is a conditioning penalty, and $C_v(S)$ is a coverage penalty that discourages selecting rows that ignore large parts of the disk or repeatedly use the same paths.

A greedy version is often more practical:

1. Start with no selected rows.

2. Add a candidate row only if it increases numerical rank outside the current polynomial space.

3. Reject rows that make $\rho_v(S)$ too large.
4. Stop when $\nu_v(S)$ is close to $p_v(r)$, or when no remaining row improves rank and coverage without damaging polynomial residuals.

This is different from using the final global diagnostics as the proof that the operator works. If one chooses rows specifically to make a global test matrix $P_d$ satisfy

$$\frac{\|AP_d\|_F}{\|P_d\|_F} \approx 0,$$

then the same quantity cannot be treated as independent evidence of success. It becomes a construction criterion. This is not wrong, but it must be reported as such.

The clean experimental separation is:

$$\begin{aligned}
&\text{during construction:} && \nu_v(S),\ \rho_v(S),\ \text{conditioning, and coverage on local disks,}\\
&\text{after construction:} && \dim \operatorname{null}(A),\ \frac{\|AP_{\text{test}}\|_F}{\|P_{\text{test}}\|_F},\ \text{and held-out synthetic graph tests.}
\end{aligned}$$

In other words, local nullity and local polynomial residuals should guide thinning. Global nullity and global residuals should remain final diagnostics, preferably evaluated on polynomial probes or synthetic examples not used to select the rows.

## 7 Example: Path Graph

Consider the path graph

$$1 - 2 - \cdots - n.$$

For an interior vertex $v = i$, a graph disk produces a left ray and a right ray. The composite geodesic through $i$ is simply an interval of the path:

$$i - q, \ldots, i, \ldots, i + q.$$

For $r = 2$, the annihilator is third order. On equally spaced vertices, a four-point window gives the familiar row

$$[-1,\ 3,\ -3,\ 1].$$

Therefore the geodesic construction reduces to ordinary one-dimensional quadratic trend filtering on path graphs, up to the boundary and window selection convention.

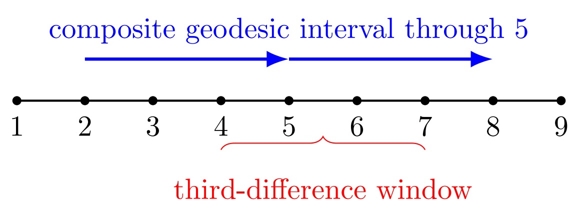


Figure 2: On a path graph, composite geodesics are ordinary intervals. The $(r + 1)$-st pathwise annihilator recovers the usual one-dimensional trend-filtering difference rows.

## 8 Example: Three-Arm Star

Now consider a three-arm star with two vertices per arm:

$$u - v_{a1} - v_{a2}, \qquad a = 1, 2, 3.$$

Composite geodesics through the center $u$ include cross-arm paths such as

$$v_{11} - u - v_{21}.$$

For $r = 1$, the annihilator is second order. With equal edge lengths, the cross-arm row is

$$f_{11} - 2f_u + f_{21}.$$

If every cross-arm second difference vanishes, then the slopes across arms are forced to be compatible. With only one vertex per arm, the null space can collapse to constants. With longer arms, the condition says something closer to: values should extend linearly through the branch point along every geodesic line that passes through it.

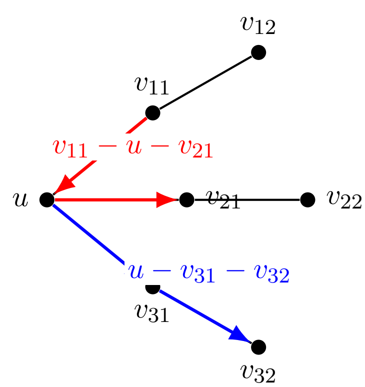


Figure 3: A star graph exposes a modeling choice. Composite geodesics include cross-arm paths through the center; arm-continuation paths are only a subset. Cross-arm rows enforce compatibility of polynomial behavior across arms.

This is not a flaw. It is a choice. The composite-geodesic construction treats two arms that form a shortest path through $u$ as two sides of a local geodesic line. If the scientific goal is branch-specific slopes, use fewer paths. If the goal is a graph analogue of line tests through a point, keep the composite geodesics.

## 9 What Should Happen on Sampled Squares?

The computational test that matters most is a graph built from points sampled from

$$[0, 1]^2.$$

Construct a symmetric $k$-nearest-neighbor graph, with edge lengths inherited from Euclidean distances. Around each vertex, choose a disk radius $\varepsilon$ and build composite geodesics.

For degree $r = 1$, the null space should approximately contain affine functions:

$$f(x, y) = a + bx + cy.$$

For degree $r = 2$, it should approximately contain quadratics:

$$f(x, y) = a + bx + cy + dx^2 + exy + gy^2.$$

For degree $r = 3$, it should approximately contain cubics:

$$\begin{aligned} f(x, y) = {} & a + bx + cy + dx^2 + exy + gy^2 \\ & + hx^3 + ix^2y + jxy^2 + ky^3. \end{aligned}$$

The dimensions in two variables are

$$\dim \mathcal{P}_r(\mathbb{R}^2) = \binom{r+2}{2}.$$

Thus the expected dimensions are:

| $r$ | polynomial space | $\dim \mathcal{P}_r(\mathbb{R}^2)$ |
|---|---|---|
| 1 | affine | 3 |
| 2 | quadratic | 6 |
| 3 | cubic | 10 |

On a finite noisy graph, the nullity of $A_{r+1,\mathrm{geo}}$ may not be exactly these numbers. A more stable diagnostic is:

$$\frac{\|A_{r+1,\mathrm{geo}} f_{\mathrm{poly}}\|_2}{\|f_{\mathrm{poly}}\|_2} \approx 0$$

for sampled polynomial signals $f_{\mathrm{poly}}$.

# 10 What Should Happen on Sampled Quadrics?

A second useful test is to sample a curved surface, for example

$$(x, y, z) = (x, y, x^2 + y^2), \qquad (x, y) \in [0, 1]^2,$$

or more generally

$$z = q(x, y).$$

Build the graph using Euclidean distances in $\mathbb{R}^3$, or intrinsic graph distances approximating the surface metric.

The expected behavior changes subtly. A function that is quadratic in the ambient coordinates $(x, y, z)$ is not the same as a quadratic in intrinsic surface coordinates. For example, the coordinate function

$$z = x^2 + y^2$$

is quadratic in the parameter coordinates $(x, y)$, but it is linear in the ambient coordinate $z$.

Therefore quadrics test two questions:

1. Does the graph operator annihilate polynomials in intrinsic local geodesic coordinates?
2. Does it behave differently from an operator built from ambient Euclidean coordinates?

This is useful because the graph construction should be metric-intrinsic. It should depend primarily on graph geodesic geometry, not on the arbitrary ambient coordinates used to store the samples.

## 11 Suggested Computational Diagnostics

For each graph and degree $r = 1, 2, 3$, build $A_{r+1,\mathrm{geo}}$ and run the following checks.

### 11.1 Polynomial Residual Tests

Sample known polynomial signals and compute:

$$\rho(f) = \frac{\|A_{r+1,\mathrm{geo}} f\|_2}{\|f\|_2}.$$

For a good degree-$r$ graph polynomial annihilator, $\rho(f)$ should be small for degree-$r$ polynomial signals and larger for degree-$(r+1)$ signals.

### 11.2 Numerical Nullity

Compute the singular values of $A_{r+1,\mathrm{geo}}$. Estimate nullity by counting singular values below a tolerance:

$$\widehat{\dim \mathrm{null}}(A) = \#\{s_i(A) : s_i(A) < \tau\}.$$

On sampled $[0, 1]^2$, compare this with

$$\binom{r+2}{2}.$$

### 11.3 Local Nullity

Because the operator is built from graph disks, global nullity can be distorted by boundaries, irregular sampling, or disconnected path families. Also compute local nullity on each disk:

$$A_{r+1,\mathrm{geo}}|_{D(v,\varepsilon)}.$$

This reveals whether each local graph disk has the right polynomial degrees of freedom.

### 11.4 Boundary Diagnostics

At boundary vertices, there may be too few balanced geodesics. The graph path Laplacian manuscript proposes mirror extensions for this problem. The same idea can be reused here: when only one geodesic ray is available, mirror the ray to create a symmetric path and construct the annihilator on the mirrored coordinate system.

## 12 Implementation Sketch

The construction can be organized as follows.

1. For each vertex $v$, compute $D(v, \varepsilon)$ and its boundary.
2. Enumerate boundary pairs $(a, b)$.
3. Keep composite paths $a \to v \to b$ that are geodesic between $a$ and $b$.
4. Assign signed path coordinates centered at $v$.

5. Build a divided-difference or weighted-regression annihilator row for degree $r$.
6. Insert the row into the global sparse matrix $A_{r+1,\mathrm{geo}}$.
7. Return diagnostics: number of disks, number of paths, boundary/mirror counts, row rank, estimated nullity, and polynomial residual tests.

# 13 Implementation Roadmap

The mathematical construction above is intentionally broader than the first computational implementation should be. The near-term implementation should therefore proceed as a sequence of auditable pieces: first build graph disks and candidate geodesic-path rows, then add local coordinate and dimension diagnostics, then use these diagnostics to thin rows and test the resulting operator.

## 13.1 Local Disk Construction

For each vertex $v$, construct a graph disk

$$D(v, \varepsilon_v) = \{u : \mathrm{dist}_G(u, v) \le \varepsilon_v\}.$$

The radius should be allowed to adapt by vertex. A useful first rule is: choose the smallest radius $\varepsilon_v$ for which the number of usable composite geodesics through $v$ reaches a requested minimum, subject to a maximum radius and a maximum number of retained paths. The implementation should record the selected radius, disk size, number of candidate composite geodesics, number of retained geodesics, and whether the maximum radius or path cap was hit.

This adaptive-radius rule is important because fixed-radius disks can be too sparse near boundaries or in locally irregular regions, while overly large disks can mix distinct geometric regimes.

## 13.2 Local Coordinate Engines

Local coordinates are not part of the definition of the pathwise annihilator, but they are useful for diagnostics, dimension estimation, and polynomial probe construction. The first implementation should support at least two local embedding backends:

`cmdscale` and `weighted_grip_edge_kk`.

The `cmdscale` backend gives a classical metric-MDS reference from local graph-geodesic distances. The weighted-GRIP $\to$ edge-KK backend should use the weighted GRIP initializer followed by edge-restricted Kamada–Kawai repair:

`grip.layout.weighted()` $\longrightarrow$ `grip.optimize.edge.kk.layout()`.

This is attractive because it attempts to realize the observed weighted graph edge lengths directly, rather than only matching all-pairs distances through a dense MDS objective.

The current weighted-GRIP $\to$ edge-KK workflow is most mature for target dimensions $1, 2, 3$. When the best three-dimensional local layout still has large edge stress, the implementation should set a diagnostic flag rather than silently treating the disk as intrinsically three-dimensional.

### 13.3 Local Dimension Estimation

The local intrinsic dimension estimate $m_v$ should be treated as an engineering guide, not as a proof of the operator's null space. Candidate dimension estimates can be obtained from several sources:

1. edge-stress elbows from local weighted-GRIP $\to$ edge-KK embeddings in dimensions $1, 2, 3$;
2. singular-value or eigenvalue elbows from local coordinates;
3. spectral diagnostics of the local graph Laplacian or local heat kernel;
4. smoothed estimates obtained by regressing raw vertexwise estimates over the graph.

The implementation should keep the individual diagnostic votes as well as the selected dimension. In particular, a local dimension cap should be visible in the output when the evidence suggests dimension greater than three.

### 13.4 Candidate Path Rows

For each disk $D(v, \varepsilon_v)$, generate candidate composite geodesic paths and convert them into $(r+1)$-st pathwise divided-difference rows. The first implementation should include:

1. a cap on the number of candidate composite geodesics per vertex;
2. random tie-breaking when multiple shortest paths exist between the same boundary pair;
3. mirror extensions at graph boundaries and in sparse one-sided local neighborhoods;
4. row-level diagnostics recording path length, endpoint distance, disk radius, and whether the row used a mirror extension.

The row-normalization rule should remain configurable because it changes the effective penalty. Important variants include no normalization, path-length normalization, disk/path-count normalization, and vertex-level normalization.

### 13.5 Row Thinning

Candidate rows can easily overconstrain dense or high-degree graph disks. Row thinning should therefore be a first-class part of the implementation rather than a post-hoc clean-up step. A useful local objective is to choose rows that keep a local polynomial design $P_v$ nearly unpenalized while still providing coverage and rank outside that polynomial space:

$$\frac{\|A_v P_v\|_F}{\|P_v\|_F} \approx 0.$$

At the same time, selected rows should improve path coverage, avoid extreme conditioning, and move local nullity toward the dimension of the desired polynomial probe space.

This means that local dimension estimates guide the thinning target, while the actual thinning criterion uses the rows themselves:

$$\dim \operatorname{null}(A_v), \qquad \frac{\|A_v P_v\|_F}{\|P_v\|_F}, \qquad \operatorname{coverage}(A_v).$$

### 13.6 Global Validation

After local rows are assembled into the global operator, the final check should be empirical. The main validation quantities are:

$$\widehat{\dim \operatorname{null}}(A), \qquad \frac{\|AP_d\|_F}{\|P_d\|_F},$$

where $P_d$ contains held-out polynomial probe signals of the target degree. For sampled two-dimensional domains, the expected polynomial dimensions are

$$3, \quad 6, \quad 10$$

for degrees $1, 2, 3$, respectively. These numbers should be treated as reference targets for well-sampled planar examples, not as universal graph truths.

### 13.7 Experimental Graph Families

The first validation suite should include graphs that isolate different failure modes:

1. path graphs, where the operator should reduce to ordinary one-dimensional divided differences;
2. star and multi-arm graphs, where branching behavior and null-space inflation are easiest to inspect;
3. sampled $[0, 1]^2$ graphs, where degree-$1, 2, 3$ polynomial nullities can be compared with $3, 6, 10$;
4. sampled quadrics over $[0, 1]^2$, which test whether the operator is intrinsic rather than an ambient-coordinate annihilator;
5. Delaunay one-skeleton graphs;
6. symmetric kNN and continuous-kNN graphs.

For each family, reports should show disk-size histograms, composite-geodesic histograms, selected-radius diagnostics, local dimension diagnostics, polynomial residuals, and numerical nullity.

## 14 Interpretation

The graph path Laplacian manuscript used composite geodesics to estimate an average of second directional derivatives. The present construction uses the same geodesic sampling idea, but changes the target:

path Laplacian: estimate and average second derivatives,

polynomial annihilator: penalize $(r + 1)$-st pathwise derivatives.

For $r = 2$, this means penalizing third pathwise derivatives. The intended effect is the graph analogue of quadratic trend filtering: functions that are locally quadratic along sampled geodesic lines should be nearly unpenalized, while deviations from local quadratic behavior should be sparse and adaptive.

## 15 Open Questions

1. Should $\Gamma(v, \varepsilon)$ include all composite geodesics, or a sampled subset for computational stability?
2. How should multiple shortest paths between the same boundary pair be handled?
3. Should path rows be normalized by path length, disk size, or number of paths through $v$?
4. Should mirror extensions be used only at graph boundaries, or also in sparse one-sided local neighborhoods?
5. On sampled $[0,1]^2$, does the estimated nullity stabilize near $3, 6, 10$ for degrees $1, 2, 3$?
6. On sampled quadrics, does the operator behave like an intrinsic polynomial annihilator rather than an ambient coordinate polynomial annihilator?

## 16 Summary

The proposed operator is a graph-intrinsic polynomial annihilator:

$$A_{r+1,\mathrm{geo}} f$$

is built by sampling composite geodesic paths through graph disks and applying $(r+1)$-st one-dimensional polynomial annihilators along those paths. It inherits the motivating idea of the graph path Laplacian, but targets graph trend filtering rather than quadratic spectral smoothing. On path graphs it reduces to ordinary one-dimensional trend-filtering differences. On richer graphs it gives a way to test whether local graph signals behave like degree-$r$ polynomials along intrinsic geodesic directions.